In [ ]:
```
📘 CUSTOMER CHURN ANALYSIS
```

In [ ]:

1. Import Libraries

In [30]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Note:

These libraries help us:

- pandas → work with data
- numpy → math operations
- matplotlib & seaborn → create graphs

2. Load Dataset

In [31]:
```
df = pd.read_csv("telecom_churn_data.csv")
df.head()
```

Out[31]:

	year	customer_id	phone_no	gender	age	no_of_days_subscribed	multi_screen	ma
0	2015	100198	409-8743	Female	36	62	no	
1	2015	100643	340-5930	Female	39	149	no	
2	2015	100756	372-3750	Female	65	126	no	
3	2015	101595	331-4902	Female	24	131	no	
4	2015	101653	351-8398	Female	40	191	no	

- This loads the dataset and shows first 5 rows.

3. Basic Information

In [32]:
```
df.shape
df.info()
df.columns
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2000 entries, 0 to 1999
Data columns (total 16 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   year                    2000 non-null   int64
 1   customer_id             2000 non-null   int64
 2   phone_no                2000 non-null   object
 3   gender                  1976 non-null   object
 4   age                     2000 non-null   int64
 5   no_of_days_subscribed   2000 non-null   int64
 6   multi_screen            2000 non-null   object
 7   mail_subscribed         2000 non-null   object
 8   weekly_mins_watched     2000 non-null   float64
 9   minimum_daily_mins      2000 non-null   float64
 10  maximum_daily_mins      2000 non-null   float64
 11  weekly_max_night_mins   2000 non-null   int64
 12  videos_watched          2000 non-null   int64
 13  maximum_days_inactive   1972 non-null   float64
 14  customer_support_calls  2000 non-null   int64
 15  churn                   1965 non-null   float64
dtypes: float64(5), int64(7), object(4)
memory usage: 250.1+ KB
```

Out[32]:

```
Index(['year', 'customer_id', 'phone_no', 'gender', 'age',
       'no_of_days_subscribed', 'multi_screen', 'mail_subscribed',
       'weekly_mins_watched', 'minimum_daily_mins', 'maximum_daily_mins',
       'weekly_max_night_mins', 'videos_watched', 'maximum_days_inactive',
       'customer_support_calls', 'churn'],
      dtype='object')
```

- shape → number of rows and columns
- info() → data types
- columns → column names

4.Drop Irrelevant Columns

In [33]:

```
df.drop(columns=['customer_id','phone_no','year'], inplace=True)
```

Reason:

- customer_id → unique identifier
- phone_no → not useful
- year → constant

5. Check Missing Values

In [34]:

```
df.isnull().sum()
```

Out[34]:
```
gender                    24
age                        0
no_of_days_subscribed      0
multi_screen               0
mail_subscribed            0
weekly_mins_watched        0
minimum_daily_mins         0
maximum_daily_mins         0
weekly_max_night_mins      0
videos_watched             0
maximum_days_inactive     28
customer_support_calls     0
churn                     35
dtype: int64
```

Note:

- This checks if any values are missing.
- If missing values exist:

In [42]:
```
# Fill categorical column with mode
df['gender'] = df['gender'].fillna(df['gender'].mode()[0])
```

In [43]:
```
# Fill numeric column with median
df['maximum_days_inactive'] = df['maximum_days_inactive'].fillna(
    df['maximum_days_inactive'].median()
)
```

In [44]:
```
# Drop rows where churn (target) is missing
df = df.dropna(subset=['churn'])
```

6. Separate Categorical & Numerical Columns

In [45]:
```
categorical_cols = df.select_dtypes(include='object').columns
numerical_cols = df.select_dtypes(exclude='object').columns

categorical_cols
numerical_cols
```

Out[45]:
```
Index(['age', 'no_of_days_subscribed', 'weekly_mins_watched',
       'minimum_daily_mins', 'maximum_daily_mins', 'weekly_max_night_mins',
       'videos_watched', 'maximum_days_inactive', 'customer_support_calls',
       'churn'],
      dtype='object')
```

Note:

- We separate text columns and number columns.

7. Summary Statistics

For Number Columns

In [47]:
```
df[numerical_cols].describe()
```

Out[47]:

	age	no_of_days_subscribed	weekly_mins_watched	minimum_daily_mins
count	1965.000000	1965.000000	1965.000000	1965.000000
mean	38.697710	99.750636	269.894275	10.198677
std	10.196352	39.887889	80.578226	2.771632
min	18.000000	1.000000	0.000000	0.000000
25%	32.000000	72.000000	218.250000	8.400000
50%	37.000000	99.000000	269.550000	10.200000
75%	44.000000	127.000000	324.000000	12.000000
max	82.000000	243.000000	526.200000	20.000000

- Shows average, minimum, maximum etc.

For categorical:

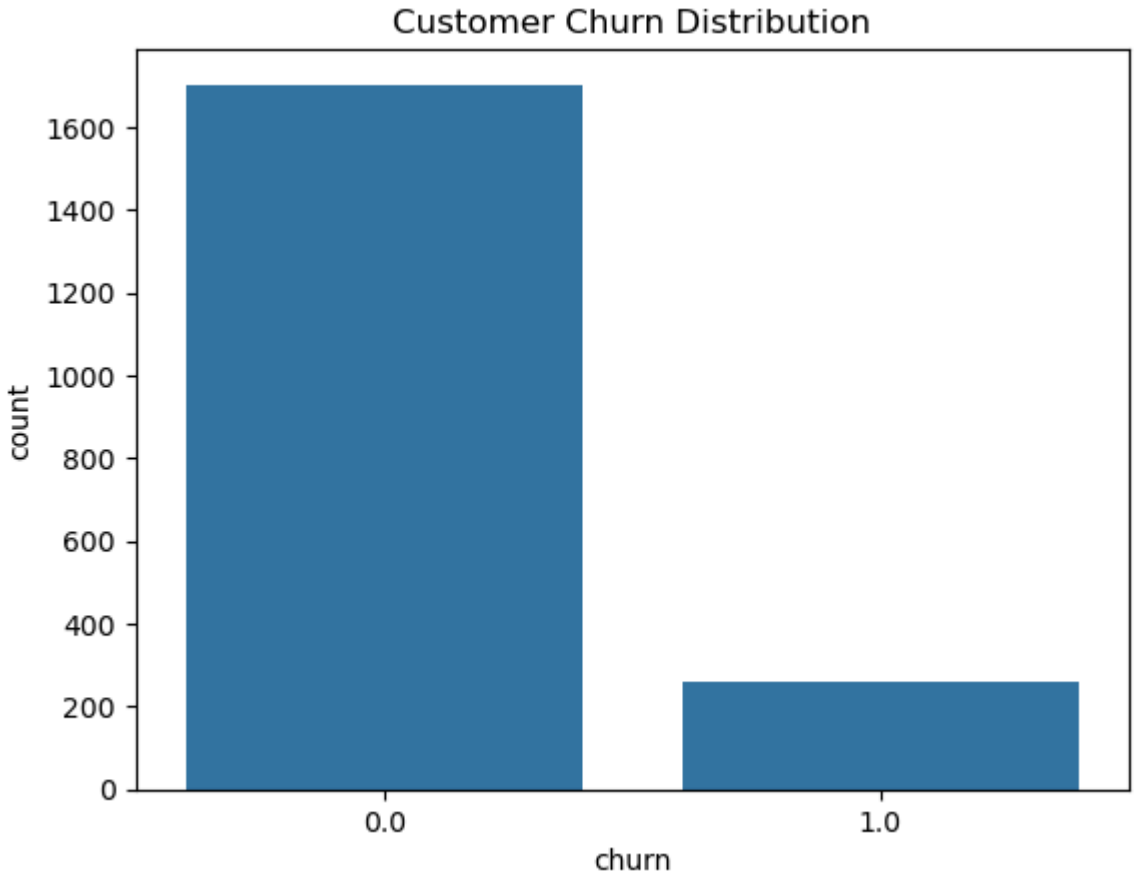
In [48]:
```
for col in categorical_cols:
    print(df[col].value_counts())
    print("-----------------")
```

```
gender
Male      1056
Female     909
Name: count, dtype: int64
-----------------
multi_screen
no     1769
yes     196
Name: count, dtype: int64
-----------------
mail_subscribed
no     1404
yes     561
Name: count, dtype: int64
-----------------
```

8. EDA Visualizations

Churn Distribution

In [49]:
```
sns.countplot(x='churn', data=df)
plt.title("Customer Churn Distribution")
plt.show()
```

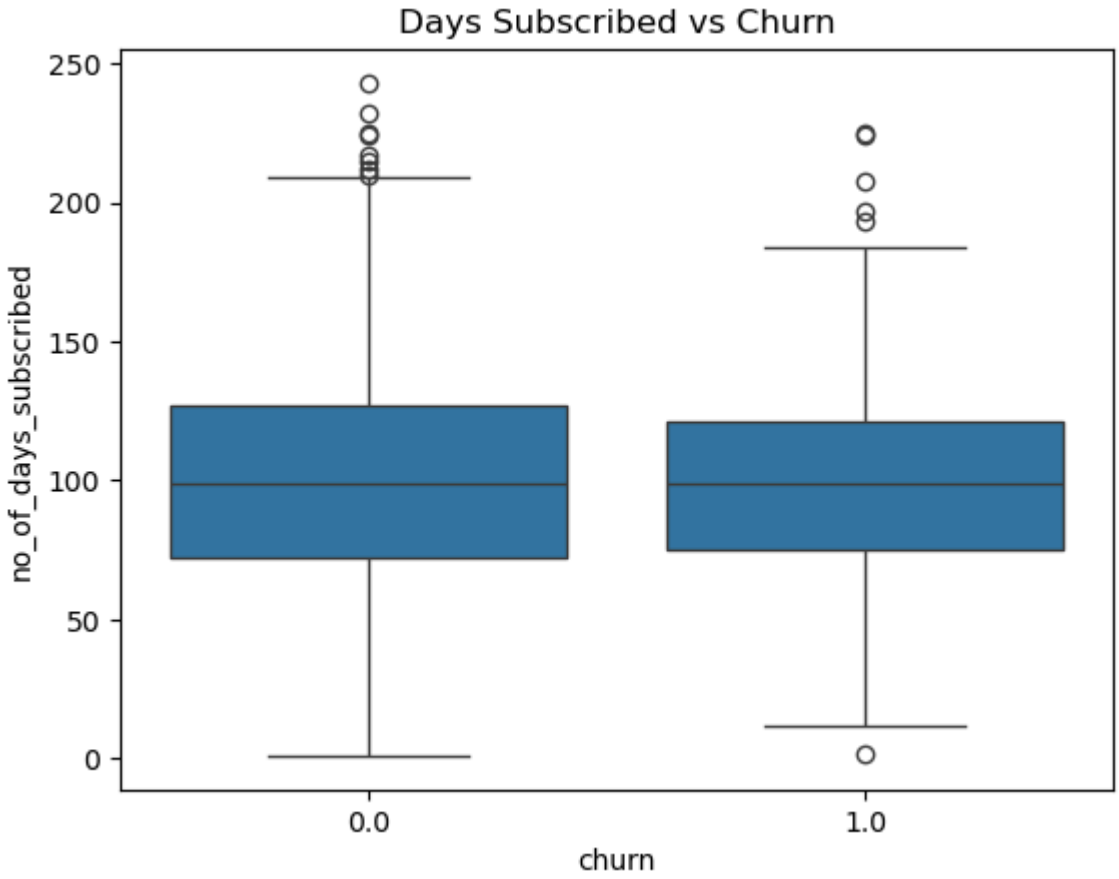


- 0 = Customer stayed
- 1 = Customer left

In [ ]:

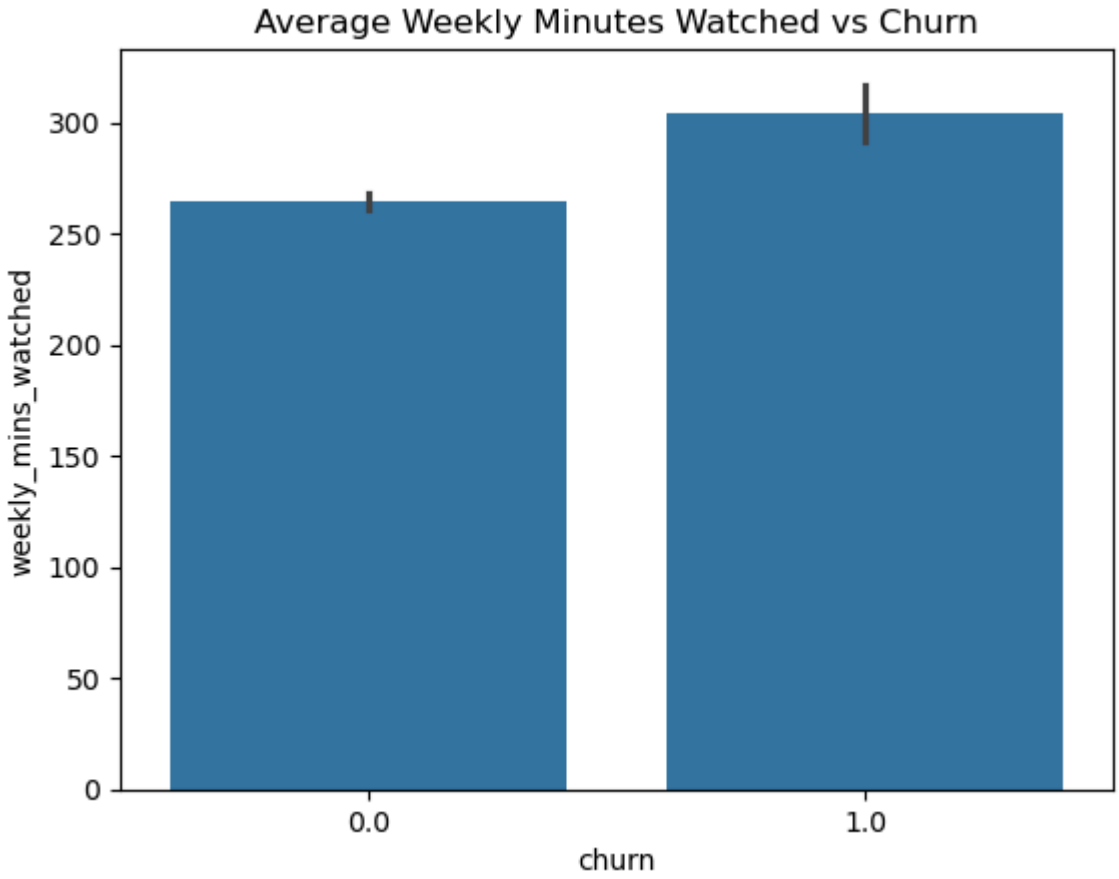
Days Subscribed vs Churn

In [50]:
```
sns.boxplot(x='churn', y='no_of_days_subscribed', data=df)
plt.title("Days Subscribed vs Churn")
plt.show()
```



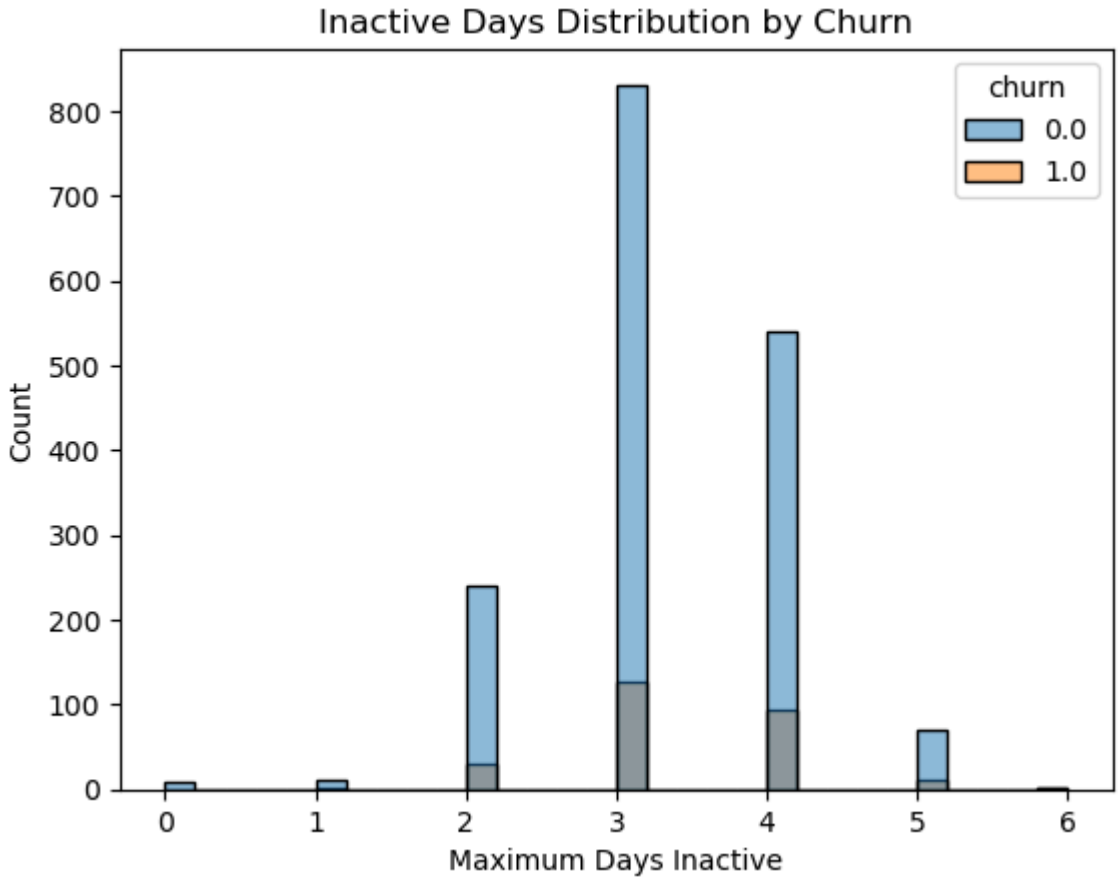
Weekly Minutes Watched vs Churn

In [51]:
```
sns.barplot(x='churn', y='weekly_mins_watched', data=df)
plt.title("Average Weekly Minutes Watched vs Churn")
plt.show()
```



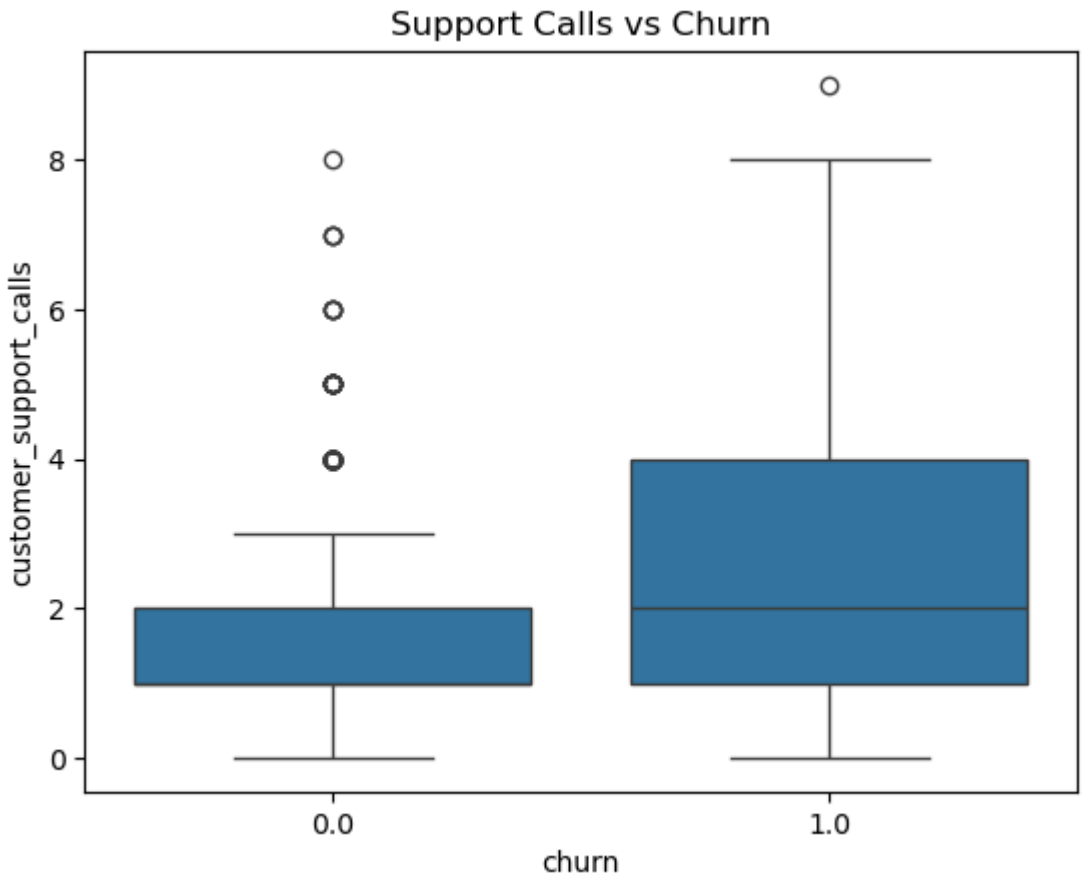
Maximum Inactive Days vs Churn (Histogram)

In [52]:
```
sns.histplot(data=df, x='maximum_days_inactive', hue='churn', bins=30)
plt.title("Inactive Days Distribution by Churn")
plt.xlabel("Maximum Days Inactive")
plt.ylabel("Count")
plt.show()
```



Support Calls vs Churn

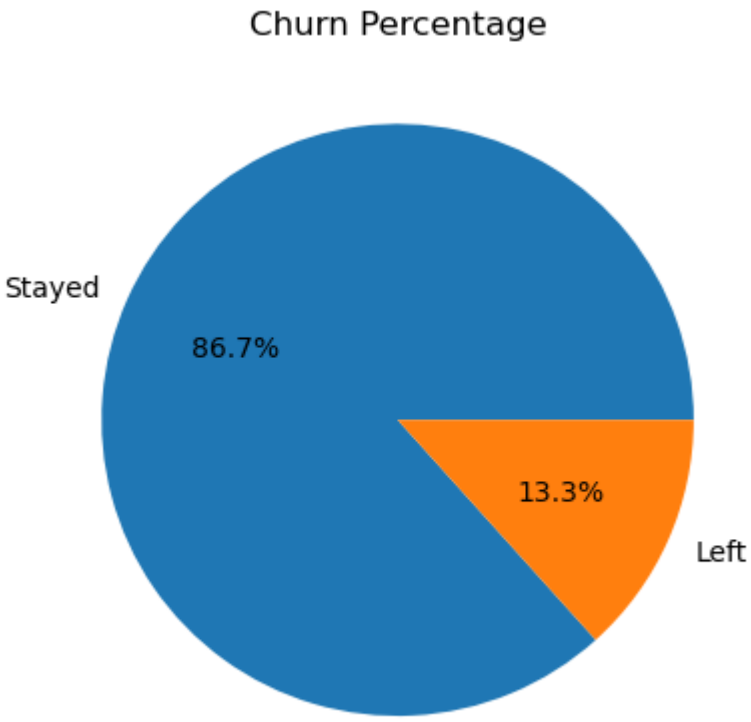
In [53]:
```
sns.boxplot(x='churn', y='customer_support_calls', data=df)
plt.title("Support Calls vs Churn")
plt.show()
```



Pie Chart

In [54]:
```
churn_counts = df['churn'].value_counts()

plt.pie(churn_counts, labels=['Stayed','Left'], autopct='%1.1f%%')
plt.title("Churn Percentage")
plt.show()
```



9. Encoding

In [55]:
```
df = pd.get_dummies(df, drop_first=True)
```

10. Model Building

In [58]:
```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
from sklearn.preprocessing import StandardScaler

# Separate features and target
X = df.drop('churn', axis=1)
y = df['churn']

# Scale features
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Train test split
X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y, test_size=0.2, random_state=42
)

# Model
model = LogisticRegression(max_iter=3000)
model.fit(X_train, y_train)

# Prediction
y_pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))
```

```
Accuracy: 0.8804071246819338
```

Final Conclusion

- The analysis shows that customer inactivity, lower subscription days, and higher support calls increase the likelihood of churn. Engagement plays a major role in customer retention. The Logistic Regression model achieved approximately 88% accuracy, indicating strong predictive performance.

In [ ]: